IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 8: Edge Detection (10 Minutes)

Tasks:

1. **Define Edge Detection in one sentence:**
   Edge detection is the process of identifying **sharp changes in intensity or brightness** in an image to locate the boundaries of objects.

2. **Name any two edge detection operators:**
   Sobel Operator and **Prewitt Operator**.

3. **Which operator gives thin and accurate edges?**
   Canny Edge Detector.

4. **Why is Gaussian smoothing used before edge detection?**
   Gaussian smoothing is used to **reduce noise and unwanted variations** in an image before detecting edges, which helps prevent false edges and improves the accuracy of edge detection.